

IT645 Lab Assignment 2

To understand and apply essential SEO techniques including on-page SEO with structured data (JSON-LD), keyword research using SemRush tool, generating XML and HTML sitemaps, and configuring robots.txt for search engine crawling. You are given a sample HTML page as the base content for this assignment. You will perform SEO improvements on this page.

A. On-page SEO with structured data using JSON-LD

- i. Identify the appropriate Schema.org type for the given content (hint: "LocalBusiness" → "BookStore").
- ii. Create a JSON-LD script containing relevant business details from the page (name, description, address, contact, opening hours, etc.).
- iii. Insert the JSON-LD script inside the <head> section of index.html.
- iv. Validate the structured data using following validator tools (insert the entire html file along with the script element):

<https://validator.schema.org/> or <https://redirection.io/tools/structured-data/test>

B. Keyword Research using SemRush Tool

- i. Explore the tool: Read about the Competitive Research and Keyword research
 - ii. Use the Keyword Magic Tool (available in free version)
 - a. Search for terms related to “book store ” and note down:
 - Top 10 keyword suggestions
 - Their search volume
 - Keyword difficulty (KD) score
 - Intent
 - iii. Use Keyword Gap Tool to compare your keyword profile with competitors and include them on your page
 - a. Add competitor store’s website domains.
 - b. Analyze their:
 - Top organic keywords
 - Opportunities (weak, missed, untapped keywords)
 - iv. Create a table listing chosen keywords, volume, intent and KD.
 - v. Explain the strategy behind choosing specific keywords you would target and why.
- ### C. Generate XML Sitemap
- i. for any website of your choice using the website <https://www.xml-sitemaps.com/>.
 - ii. For the Bookstore website, create:

- a. XML Sitemap (sitemap.xml) manually
 - b. HTML Sitemap (sitemap.html)
- iii. Ensure it includes:
 - a. Home page (index.html)
 - b. Menu page (can be dummy: menu.html)
 - c. Contact page (contact.html)
- iv. Place sitemap.xml and sitemap.html in the root directory.
- v. Validate XML sitemap using:

<https://www.xml-sitemaps.com/validate-xml-sitemap.html>
- D. Create robots.txt in the root directory.
 - i. Include the xml sitemap in robots.txt
 - ii. Allow search engines to crawl all pages except:
 - a. Admin pages (/admin/)
 - b. Test pages (/test/)
- E. Web page performance optimization
 - iii. Create an HTML page with book images using flexbox layout. Add a video about the book collection.
 - iv. Perform optimizations: Perform compression of images using online image compression tools such as compression.io, enable lazy loading of images and video.
 - a. Run the optimized page through the Google PageSpeed Insights (pagespeed.web.dev) and Compare the core web vitals: LCP, INP and CLS before and after optimization.

Submit:

1. Updated index.html with JSON-LD structured data.
2. Screenshots of validation results
3. SemRush keyword research table and strategy.
4. XML and HTML sitemap files.
5. robots.txt file.
6. Performance optimization report